

LAB # 1: INTRODUCTION TO MS ACCESS (TABLES, FORMS)

Objective:

- 1-Introduction to MS ACCESS
- 2-How to create tables.
- 3-How to create forms.

Scope:

The student should know the following:

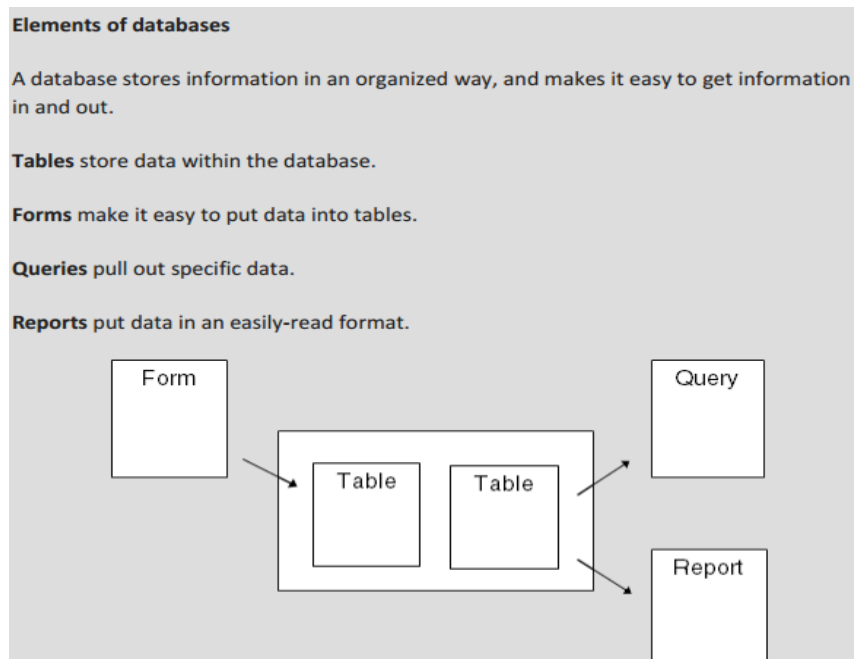
1. MS Access basics.
2. MS Access Practice.
3. Basic exercises.

Useful Concepts:

Table name, its column name and column's datatypes. How to create queries on tables.
Forms and Reports.

Discussion:

Queries, Reports, Forms.



Understanding Access Databases

As you already know, a database is a collection of information. In Access, every database is stored in a single file. That file contains *database objects*, which are the components of a database.

Database objects are the main players in an Access database. Altogether, you have six different types of database objects:

- ☐ **Tables** store information.
- ☐ **Queries** let you quickly perform an action on a table.
- ☐ **Forms** Forms provide an easy way to view or change the information in a table.
- ☐ **Reports** help you print some or all of the information in a table.
- ☐ **Macros** are mini programs that automate custom tasks.
- ☐ **Modules** are files that contain Visual Basic code.

CREATE A BLANK NEW DATABASE:

1. Start Access.

Here you can create a new database or open an existing one.

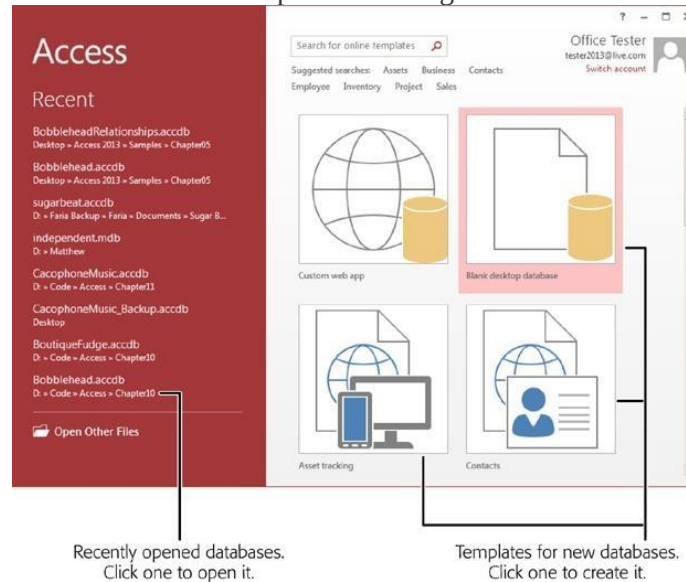


Figure 1-1. When you start Access, you see this two-part welcome page. On the left is a list of recently opened databases (if you have any). On the right is a list of templates that you can use to create a new database.

2. Click the “Blank desktop database” template.

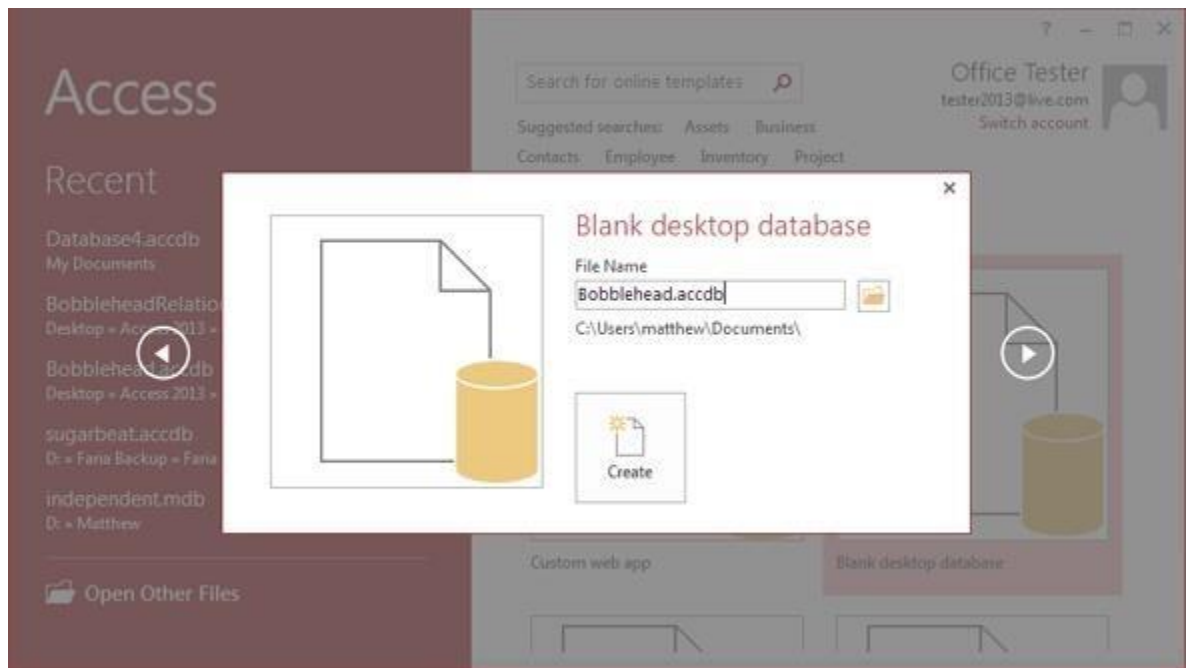
When you choose to create a blank database, that is exactly what you get—a new, empty database file with no tables or other database objects.

Other templates let you create databases that are preconfigured for specific scenarios and certain types of data. The cool sounding “Custom web app” template is a special case. It lets you create a web-enabled database that runs on SharePoint.

No matter which template you click, Access pops open a new window that lets you choose a name and location for your new database.

3. Type a file name for the database you are about to create.

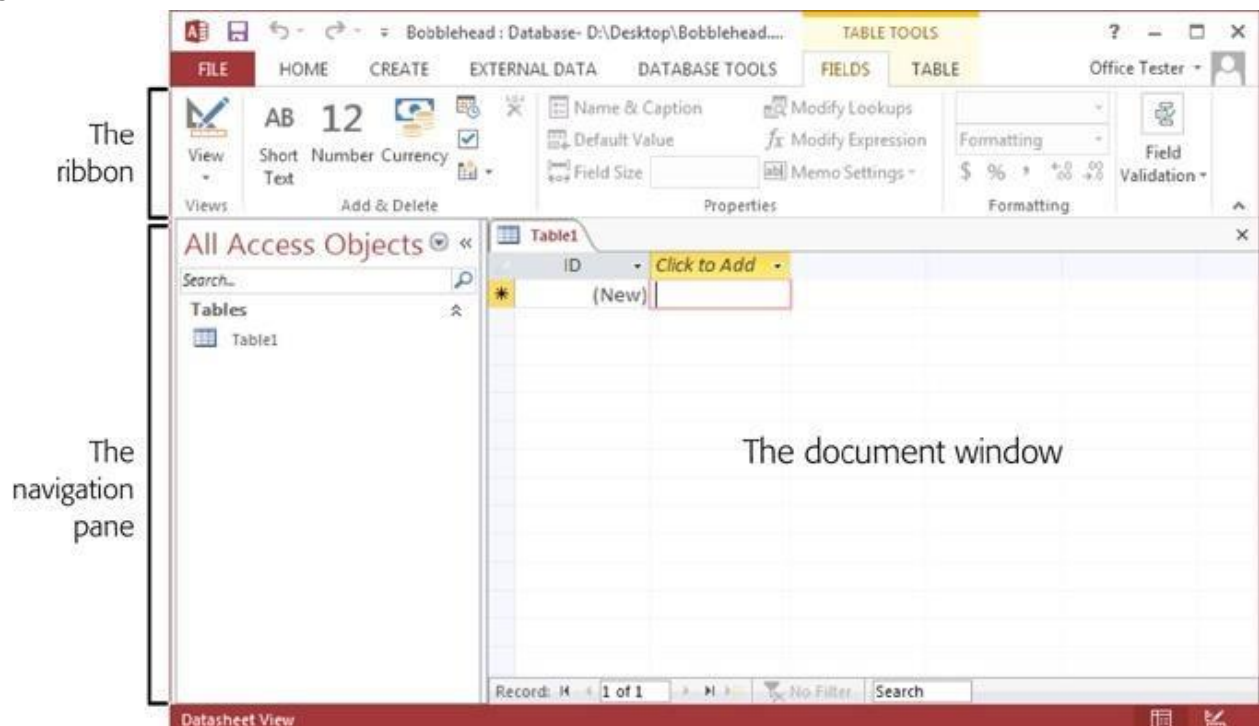
Access stores all the information for a database in a single file with the extension `.accdb` (which stands for “Access database”).



4. Click the big Create button (under the File Name box).

Access creates your database file and then shows a datasheet where you can get to work creating your first table.

3



BUILDING YOUR FIRST TABLE

Tables are information containers. Every database needs at least one table—without it, you can't store any data. In a simple database, like the Bobblehead database, a single table (which we'll call Dolls) is enough. But if you find yourself wanting to store several lists of related information, you need more than one table.

The name of the table

A field named Character

A record

ID	Character	Manufacturer	PurchasePrice	DateAcquired
1	Homer Simpson	Fictional Industries	\$7.99	1/1/2013
2	Edgar Allan Poe	Hobergarten	\$14.99	1/30/2013
3	Frodo	Magiker	\$8.95	2/4/2013
4	James Joyce	Hobergarten	\$14.99	3/3/2013
5	Jack Black	All Dolled Up	\$3.45	3/3/2013
* (New)				

Record: 1 of 7

Figure 1-5. In a table, each record occupies a separate row. Each field is represented by a separate column. In this table, it's clear that you've added five bobblehead dolls. You're storing information for each doll in five fields (ID, Character, Manufacturer, PurchasePrice, and DateAcquired).

Before you start designing this table, you need to know some very basic rules:

- **A table is a group of records.** A record is a collection of information about a single thing. In the *Dolls* table, for example, each record represents a single bobblehead doll. In a *Family* table, each record would represent a single relative. In a *Products* table, each record would represent an item that's for sale. You get the idea. When you create a new database, Access starts you out with a new table named *Table1*, although you can choose a more distinctive name when you decide to save it.
- **Each record is subdivided into fields.** Each field stores a distinct piece of information. For example, in the *Dolls* table, one field stores the person on whom the doll is based, another field stores the price, another field stores the date you bought it, and so on.
- **Tables have a rigid structure.** In other words, you can't bend the rules. If you create four fields, every record must have four fields (although it's acceptable to leave some fields blank if they don't apply).
- **Newly created tables get an ID field for free.** The ID field stores a unique number

for each record. (Think of it as a reference number that will let you find a specific record later on.) The best part about the ID field is that you can ignore it when you're entering a new record. Access chooses a new ID number for you and inserts it in the record automatically.

Creating a Simple Table

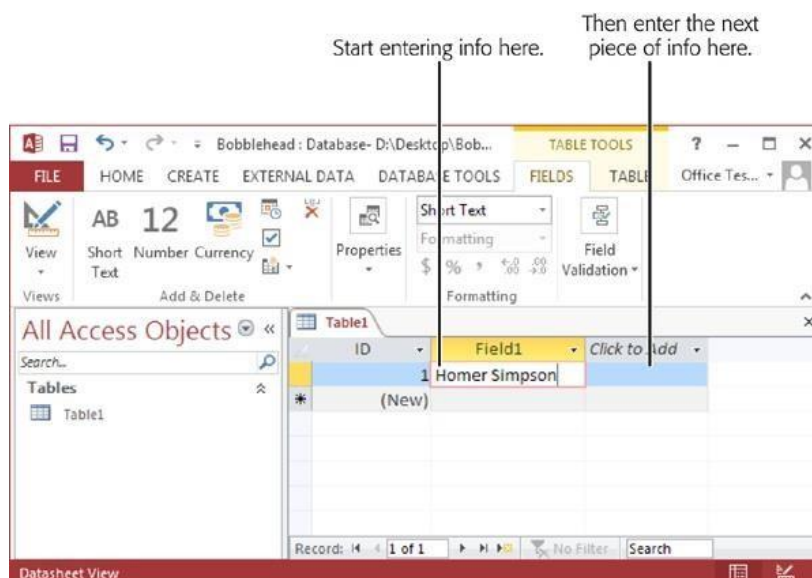
When you first create a database, it's almost empty. But to get you started, Access creates your first database object—a table named Table1. The problem is, this table begins life completely blank, with no defined fields (and no data).

You can customize a table in two ways:

- **Design view** lets you precisely define all aspects of a table before you start using it.
- **Datasheet view** is where you enter data into a table. Datasheet view also lets you build a table on the fly as you insert new information.

The following steps show you how to turn a blank new table (like Table1) into the Dolls table by using the Datasheet view:

1. **To define your table, simply add your first record.**
2. **In the datasheet's rightmost column, under the "Click to Add" heading, type the first piece of information for the record (see Figure 1-6).**



3. Press Tab to move to the next field, and return to step 2

Note: You may notice one quirk—a harmless one—when you add your first record. As you add new fields, Access may change the record's ID value of the record (changing it from 1 to 2 to 3, for example). Because the new record hasn't been inserted yet, every time you change the table's design by adding a new field, Access starts the process over and picks a new ID number, just to be safe. This automatic renumbering doesn't happen if you officially add the record (say, by moving down to the next row, or, in the ribbon, by clicking Home→Records→Save) and then add more fields to the table. However, there's really no reason to worry about the ID number. As long as it's unique—and Access guarantees that it is—the exact value is unimportant.

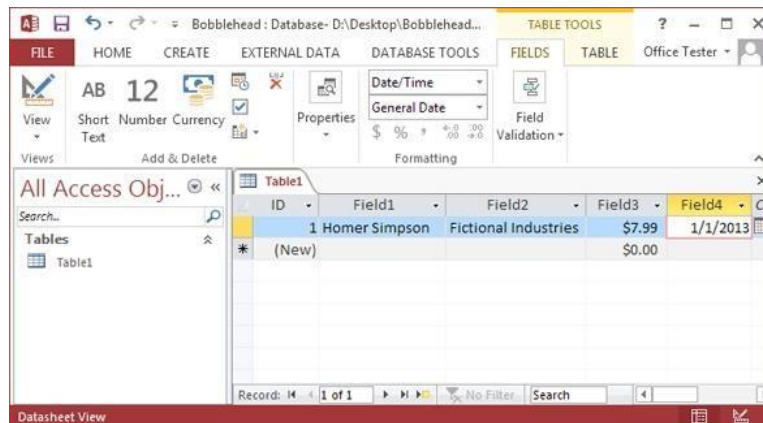


Figure 1-7. The only problem with this example so far is that as you enter a new record, Access creates spectacularly useless field names. You see its choices at the top of each column (they have names like Field1, Field2, Field3, and so on). The problem with using these meaningless names is that they may lead you to enter a piece of information in the wrong place. You could all too easily put the purchase price in the date column.

4. It's time to fix your column names. Double-click the first column title (like Field1).

5. Type a new name, and then press Enter.

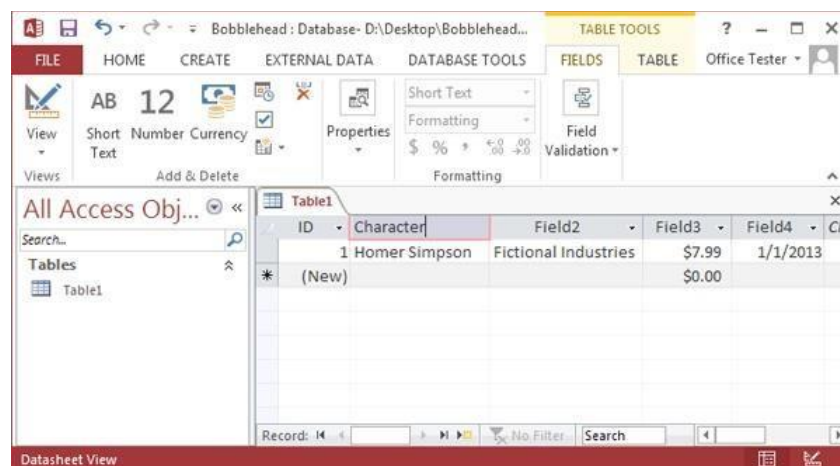


Figure 1-8. To specify better field names, double-click the column title. Next, type the real fieldname, and then press Enter. Page 90 has more about field naming, but for now just stick to short, text-only titles that don't include any spaces, as shown here.

6. Press Ctrl+S or choose File→Save to save your table.

7. Type a suitable table name, and then click OK.

Congratulations! The table is now a part of your database.

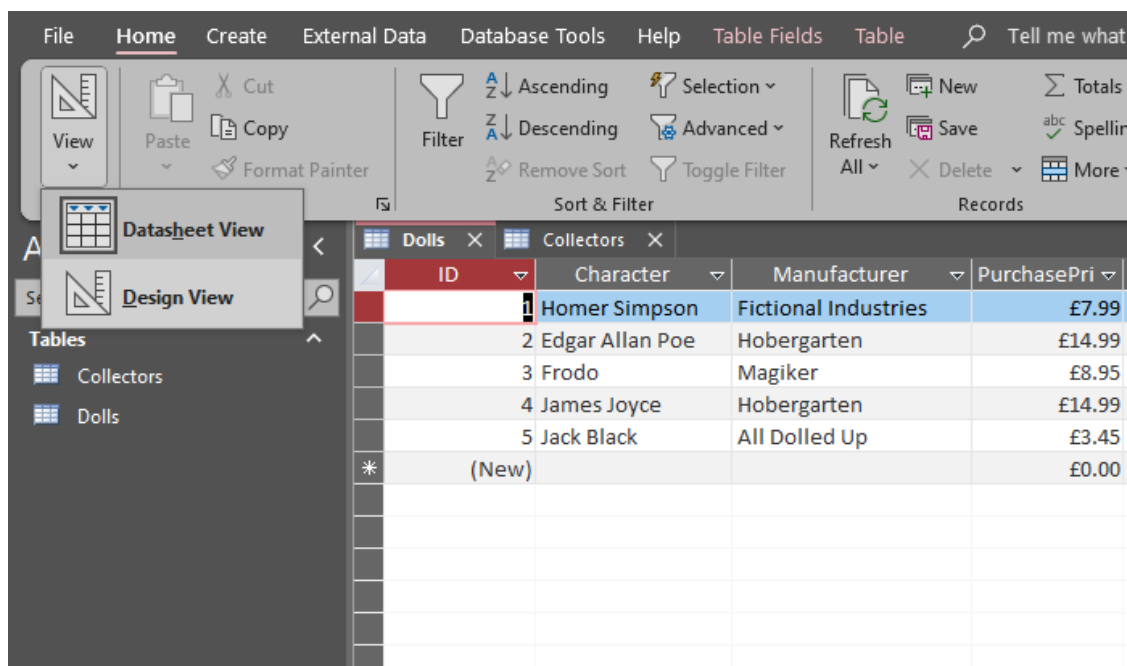
Editing a Table

To fill the Dolls table, you use the same datasheet you used to define the table. You can perform three basic tasks:

1. **Editing a record.** Move to the appropriate spot in the datasheet (using the arrowkeys or the mouse), and then type in a replacement value.
2. **Inserting a new record.** Move down to the bottom of the table to the row that has an asterisk (*) on the left. This row doesn't actually exist until you start typing some information. At that point, Access creates the row and moves the asterisk down to the next row. You can repeat this process endlessly to add as many rows as you want (Access can handle millions).
3. **Deleting a record.** You have several ways to remove a record, but the easiest is to right-click the margin immediately to the left of the record, and then choose Delete Record.

Adding Fields

Open the Design View from the bottom left of your screen or via the view column in the Home Section.



Design View

You can Add new fields or change their Data Type from the Design View Like so.

Field Name	Data Type
ID	AutoNumber
Character	Short Text
Manufacturer	Short Text
PurchasePrice	Currency
DateAcquired	Date/Time

In order to edit a field's properties just left click on the desired field and its properties will be displayed automatically.

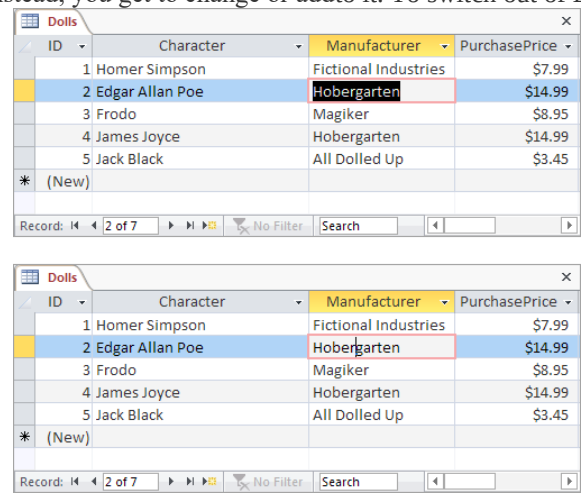
Field Name	Data Type
Character	Short Text
Manufacturer	Short Text
PurchasePrice	Currency
DateAcquired	Date/Time

General	
Format	Currency
Decimal Places	Auto
Input Mask	
Caption	
Default Value	0
Validation Rule	
Validation Text	
Required	No
Indexed	No
Text Align	General

Edit Mode

You’ll probably spend a lot of time working with the datasheet. So, settle in. To make your life easier, it helps to understand a few details.

Note: As you already know, you can use the arrow keys to move from field to field or row to row. However, you may have a bit of trouble editing a value. When you start typing, Access erases any existing content. To change this behavior, you need to switch into *Edit mode* by pressing F2; in Edit mode, your typing doesn’t delete the stuff that’s already in that field. Instead, you get to change or add to it. To switch out of Edit mode, you press F2 again.



Datasheet Shortcut Keys

Power users know the fastest way to get work done is to use tricky keyboard combinations like Ctrl+Alt+Shift+*. Although you can’t always easily remember these combinations, a couple of tables can help you out. [Table 1-1](#) lists some useful keys that can help you whiz around the datasheet.

Table 1-1. Keys for Moving Around the Datasheet

KEY	RESULT
Tab (or Enter)	Moves the cursor one field to the right, or down when you reach the edge of the table. This key also turns off Edit mode if it’s currently switched on.

KEY	RESULT
Shift+Tab	Moves the cursor one field to the left, or up when you reach the edge of the table. This key also turns off Edit mode.
→	Moves the cursor one field to the right (in Normal mode), or down when you reach the edge of the table. In Edit mode, this key moves the cursor through the text in the current field.
←	Moves the cursor one field to the left (in Normal mode), or up when you reach the edge of the table. In Edit mode, this key moves the cursor through the text in the current field.
↑	Moves the cursor up one row (unless you're already at the top of the table). This key also turns off Edit mode.
↓	Moves the cursor down one row (or it moves you to the "new row" position if you're at the bottom of the table). This key also turns off Edit mode.
Home	Moves the cursor to the first field in the current row. This key brings you to beginning of the current field if you're in Edit mode.
End	Moves the cursor to the last field in the current row. This key brings you to the end of the current field if you're in Edit mode.
Ctrl+Home	Moves the cursor to the first field in the first row. This key doesn't do anything if you're in Edit mode.
Ctrl+End	Moves the cursor to the last field in the last row. This key doesn't do anything if you're in Edit mode.

Table 1-2 lists some convenient keys for editing records.

Table 1-2. Keys for Editing Records

KEY	RESULT
Esc	Cancels any changes you've made in the current field. This key works only if you use it in Edit mode. Once you move to the next cell, the change is applied. (For additional cancellation control, try the Undo feature, described next.)
Ctrl+Z	Reverses the last edit. Unfortunately, the Undo feature in Access isn't nearly as powerful as it is in other Office programs. For example, Access lets you reverse only one change, and if you close the datasheet, you can't even do that. You can use Undo right after you insert a new record to remove it, but you can't use the Undo feature to reverse a delete operation.

KEY	RESULT
Ctrl+”	Copies a value from the field that’s immediately above the current field. This trick is handy when you need to enter a batch of records with similar information. Figure 1-11 shows this often-overlooked trick in action.
Ctrl+;	Inserts today’s date into the current field. The date format is based on computer settings, but expect to see something like “12-24-2013.” You’ll learn more about how Access works with dates on Date/Time .
Ctrl+Alt+Space	Replaces whatever value you’ve entered with the field’s default value. You’ll learn how to designate a default value on Setting Default Values .

Cut, Copy, and Paste

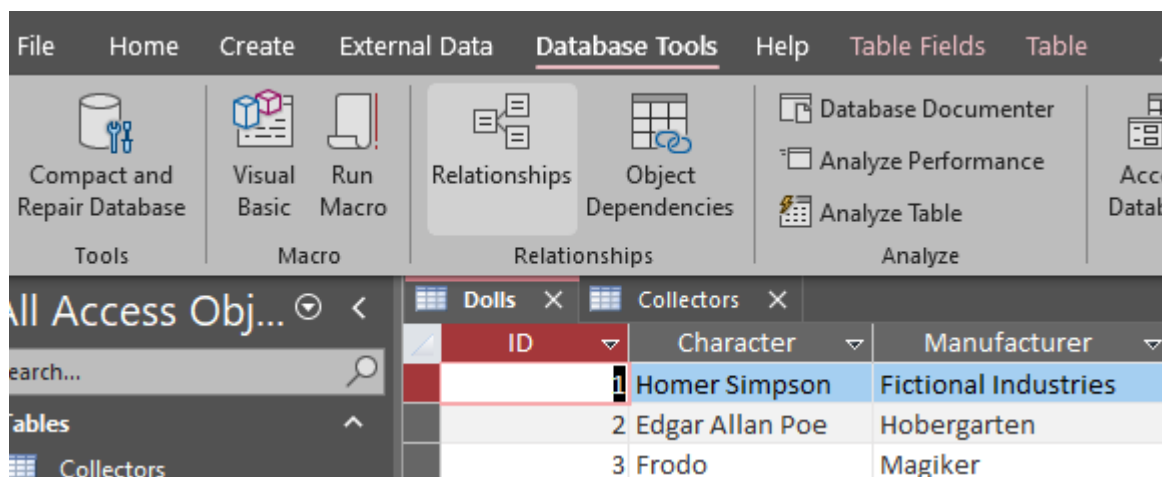
Access, like virtually every Windows program, lets you cut and paste bits of information from one spot to another. This trick is easy using just three shortcut keys: Ctrl+C to copy, Ctrl+X to cut (similar to copy, but the original content is deleted), and Ctrl+V to paste. When you’re in Edit mode, you can use these keys to copy whatever you’ve selected. If you’re not in Edit mode, the copying or cutting operation grabs all the content in the field.

Shrinking a Database

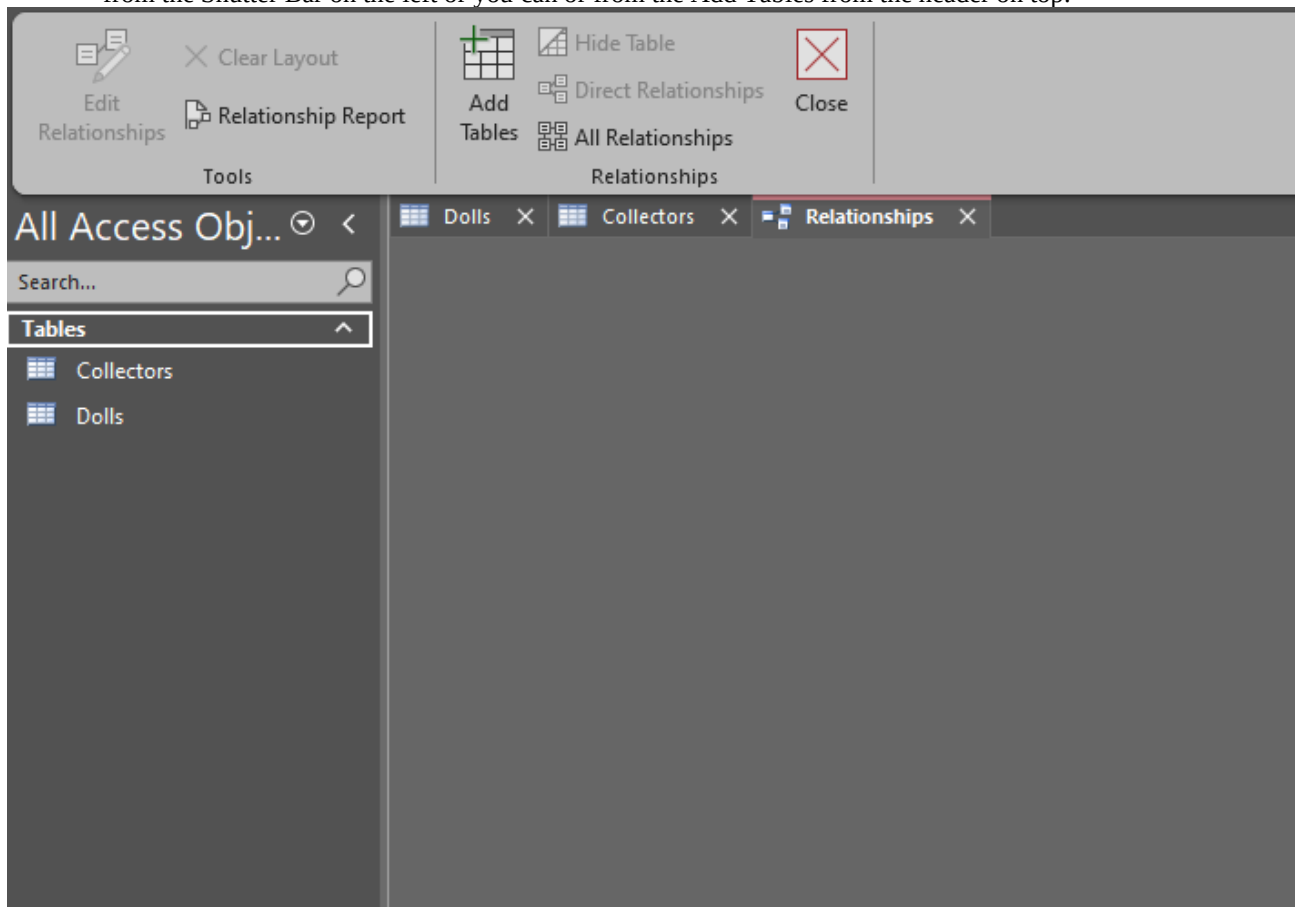
After you’ve been working with a database for a while, you might notice that its size bloats up like a week-old fish in the sun. If you want to trim your database back to size, you can use a feature called *compacting*. To do so, just choose File→Info and click the big Compact & Repair Database button. Access then closes your database, compacts it, and opens it again. If it’s a small database, these three steps unfold in seconds. The amount of space you reclaim varies widely, but it’s not uncommon to have a 20 MB database shrink down to a quarter of its size.

RELATIONSHIPS

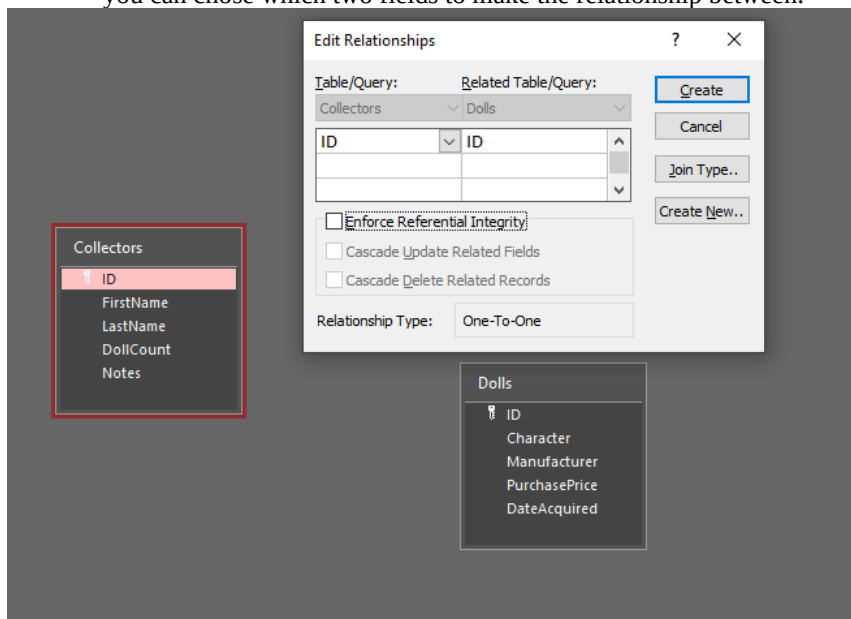
- Open the Relationships tab from the Database Tools Section.



- Now you can add the necessary tables you want to make relationships between either by drag and drop from the Shutter Bar on the left or you can or from the Add Tables from the header on top.



- To make a relationship between two fields you just have to left click on the first attribute and then drag on the second field you want to make the relationship with. After this a pop up will open through which you can chose which two fields to make the relationship between.



- Click on the Create button and you're done.

FORMS

There's a lot you can do design-wise with forms in Microsoft Access. You can create two basic types of forms –

- Bound forms
- Unbound forms

Bound Forms

Bound forms are connected to some underlying data source such as a table, query, or SQL statement.

Unbound Forms

- These forms are not connected to an underlying record or data source.
- Unbound forms could be dialog boxes, switch boards, or navigation forms.

Types of Bound Forms

There are many types of bound forms you can create in Access. Let us understand the types –

Single Item Form

This is the most popular one and this is where the records are displayed — one record at a time.

Multiple Item Form

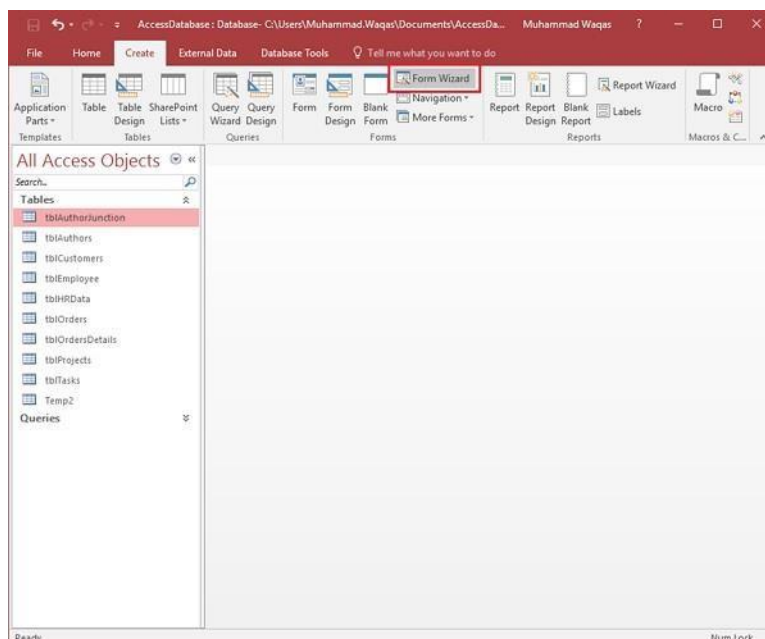
This displays multiple records at a time from that bound data source.

Split Form

The form is divided into halves, either vertically or horizontally. One half displays a single item or record, and the other half displays a list or provides a datasheet view of multiple records from the underlying data source.

Creating Forms

There are a few methods you can use to create forms in Access. For this, open your Database and go to the **Create** tab. In the Forms group, in the upper right-hand corner you will see the Form Wizard button.

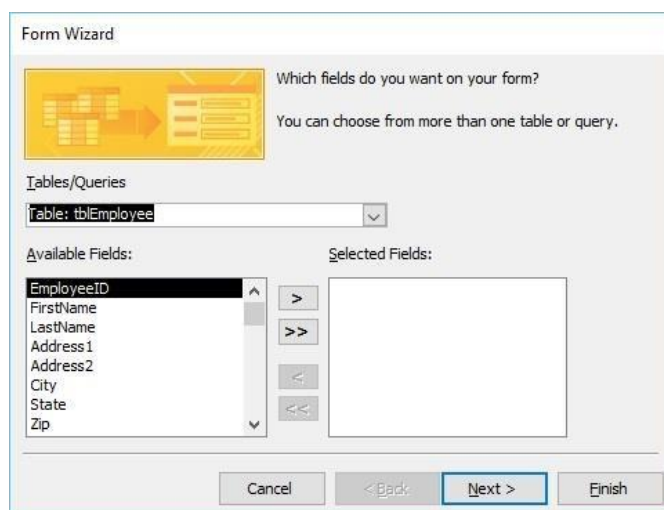


Click on that button to launch the Form Wizard.

On this first screen in the wizard, you can select fields that you want to display on your form, and you can choose from fields from more than one table or a query.

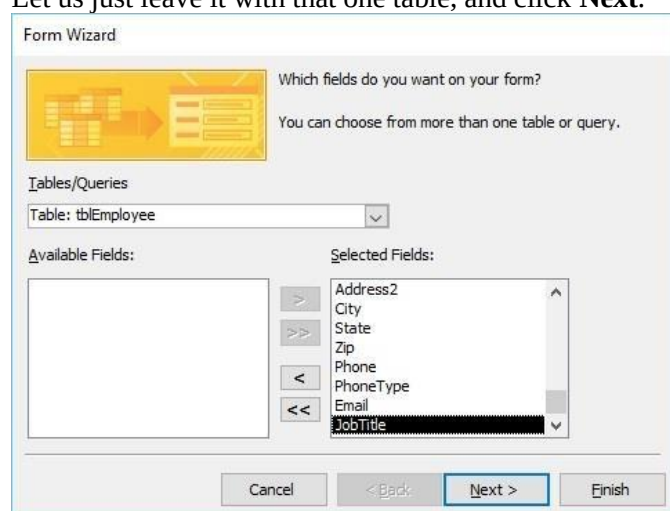
Let us assume we want to simply have a quick form that we are going to use for data entry for our employee information.

From **Tables/Queries** drop-down list, select **tblEmployees** table. Click on the double arrow to move all the fields at once.



The screenshot shows the 'Form Wizard' dialog box. At the top, it asks 'Which fields do you want on your form?' and states 'You can choose from more than one table or query.' Below this, the 'Tables/Queries' dropdown menu is set to 'Table: tblEmployee'. Under 'Available Fields:', a list of fields is shown: EmployeeID, FirstName, LastName, Address1, Address2, City, State, and Zip. The 'Selected Fields:' box is currently empty. Navigation buttons at the bottom include 'Cancel', '< Back', 'Next >', and 'Finish'.

Let us just leave it with that one table, and click **Next**.



The screenshot shows the 'Form Wizard' dialog box at the next step. The 'Tables/Queries' dropdown remains 'Table: tblEmployee'. The 'Available Fields:' box is now empty. The 'Selected Fields:' box now contains a list of fields: Address2, City, State, Zip, Phone, PhoneType, Email, and JobTitle. The 'Next >' button is highlighted in blue.

The following screen in the Form Wizard will ask for the layout that we would like for our form. We have **columnar**, **tabular**, **datasheet** and **justified** layouts. We will choose the columnar layout here and then click **Next**.

Form Wizard

What layout would you like for your form?



☒ Columnar

☐ Tabular

☐ Datasheet

☐ Justified

Cancel < Back Next > Finish

In the following screen, we need to give a title for our form. Let us call it **frmEmployees**.

Form Wizard



What title do you want for your form?

That's all the information the wizard needs to create your form.

Do you want to open the form or modify the form's design?

☒ Open the form to view or enter information.

☐ Modify the form's design.

Now, take a look at the following screenshot. This is what your form looks like. This is a single item form, meaning one record is displayed at a time and further down you can see the navigation buttons, which is telling us that this is displaying the record 1 of 9. If you click on that button then, it will move to the next record.

The screenshot shows the Microsoft Access 2007 interface. The title bar indicates the file path is 'C:\Users\Muhammad\My Documents\Access\Access2007_2\Access2007_2.accdb'. The ribbon is set to 'Form' and 'Create'. The 'All Access Objects' task pane on the left shows 'frmEmployee' selected under 'Forms'. The main window displays the 'frmEmployee' form with the following fields and values:

Field Name	Value
Employee ID	1
FirstName	Max
LastName	Clay
Address1	2004 Muhawe St
Address2	Optional
City	Schramberg
State	IL
Zip	60234
Phone	(847) 555-6492
Phone Type	Home
Email	mclay@mycompany.com
JobTitle	Accounting Assistant

If you want to jump to the very last record in that form or that table, you can use the button right beside that right arrow, the arrow with a line after it, that's the last record button. If you want to add new employee information, go to the end of this records and then after 9 records you will see a blank form where you can begin entering out the new employee's information.

The top screenshot shows the 'frmEmployee' form in Microsoft Access. The form contains the following fields and values:

Field	Value
Employee ID	3
FirstName	Jamesl
LastName	Frank
Address1	6433 Morgan Ln
Address2	Optional
City	Schaumburg
State	IL
Zip	60193
Phone	(224) 555-6631
Phone Type	Home
Email	jfrank@mycompany.com
JobTitle	Accounting Manager

The bottom screenshot shows the 'frmEmployee' form with a new record being added. The Employee ID field is highlighted, and the word 'New!' appears in the First Name field.

Field	Value
Employee ID	New!
FirstName	
LastName	
Address1	
Address2	
City	
State	
Zip	
Phone	
Phone Type	
Email	
JobTitle	

This is one example of how you can create a form using the Form Wizard. Let us now close this form and go to the Create tab. After you are done making a form you can edit it's layout by going into the Layout mode or its design by going into the Design View from the bottom right of your window or from the **Views** Section in the **Home Layout** on top. The Design View gives you much more detailed aspects of the structure(Header, Footer).

Now forms can also be use to insert records or edit already available data in a Table. You can toggle through each record from the arrow keys at the bottom left of the form. You can also search for a specific string throughout the records of the Table.

The screenshot shows the Microsoft Access interface with a form open for the 'Dolls' table. The form contains the following data:

Field	Value
ID	
Character	Homer Simpson
Manufacturer	Fictional Industries
PurchasePrice	£7.99
DateAcquired	01/01/2013

The bottom status bar indicates 'Record: 14 of 5' and includes a search bar.

Now we will create a slightly more complicated form using Wizard. Click the Form Wizard and this time, we will choose fields from a couple of different tables

In this Form Wizard, let us choose **tblProjects** for **Tables/Queries**, and select a few Available Fields such as ProjectID, ProjectName, ProjectStart, and ProjectEnd. These fields will now move to Selected Fields.

The screenshot shows the Microsoft Access Form Wizard dialog box. The 'Tables/Queries' dropdown is set to 'Table: tblProjects'. The 'Available Fields' list includes the following fields:

- Contracts.FileName
- Contracts.FileTimeStamp
- Contracts.FileType
- Contracts.FileURL
- Budget
- ProjectNotes
- OutOfPrint
- DateAdded

The 'Selected Fields' list includes the following fields:

- ProjectID
- ProjectName
- ProjectStart
- ProjectEnd

The 'Next >' button is highlighted.

Now select **tblTasks** for Tables/Queries and send over the TaskID, ProjectID, TaskTitle, StartDate, DueDate and PercentComplete. Click **Next**.

Form Wizard

Which fields do you want on your form?
You can choose from more than one table or query.

Tables/Queries
Table: tblTasks

Available Fields:

- Description
- Attachments
- Attachments.FileData
- Attachments.FileFlags
- Attachments.FileName
- Attachments.FileTimeStamp
- Attachments.FileType
- Attachments.FileURL

Selected Fields:

- ProjectStart
- ProjectEnd
- TaskID
- tblTasks.ProjectID
- TaskTitle
- StartDate
- DueDate
- PercentComplete

Cancel < Back Next > Finish

Form Wizard

How do you want to view your data?

by tblProjects
by **tblTasks**

tblProjects_ProjectID, ProjectName, ProjectStart, ProjectEnd, TaskID, tblTasks_ProjectID, TaskTitle, StartDate, DueDate, PercentComplete

☒ Single form ☐ Linked forms

Cancel < Back Next > Finish

Here, we want to retrieve data from a couple of different objects. We can also choose from options on how we want to arrange our form. If we want to create a flat form, we can choose to arrange by **tblTasks**, which will create that single form, with all the fields laid out in flat view as shown above.

However, if we want to create a hierarchical form based on that one-to-many relationship, we can choose to arrange our data by tblProjects.

Form Wizard

How do you want to view your data?

by **tblProjects**
by tblTasks

tblProjects_ProjectID, ProjectName, ProjectStart, ProjectEnd

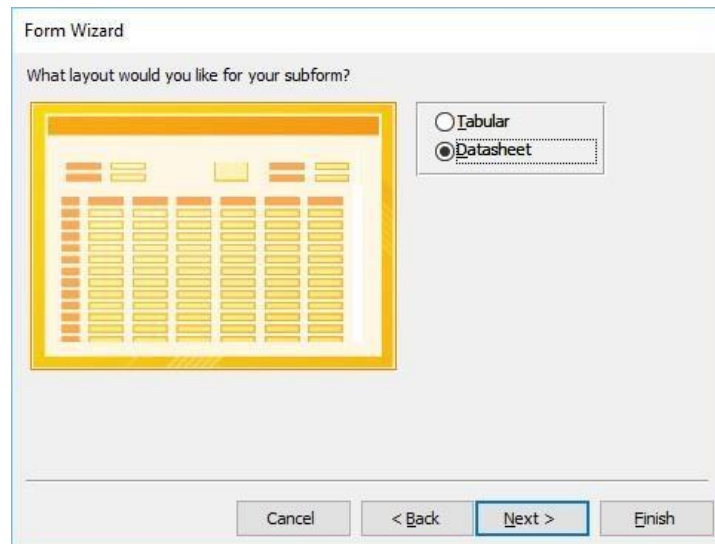
TaskID, tblTasks_ProjectID, TaskTitle, StartDate, DueDate, PercentComplete

☒ Form with subform(s) ☐ Linked forms

Cancel < Back Next > Finish

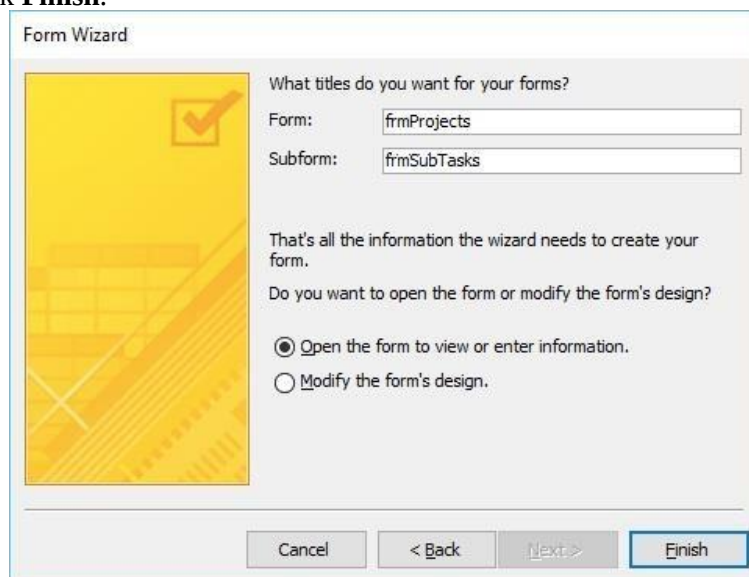
In the above window, we have the option to include a **subform** for **tblTasks**, or we can make that a linked form. This linked form is where tblProjects will have a button that will launch that second form filtered to the project that we have selected in that underlying projects form. Let us now select the **Form with subform(s)**, and then click **Next**.

In the following screen, you can choose a layout for your subform. The Datasheet View gets selected by default. The Datasheet View is similar to Table View. Now, click **Next**.



The 'Form Wizard' dialog box is shown with the title 'Form Wizard'. Below the title bar, the text 'What layout would you like for your subform?' is displayed. On the left, there is a preview of a form layout with a yellow border and a grid of fields. On the right, there are two radio button options: 'Tabular' and 'Datasheet'. The 'Datasheet' option is selected, indicated by a filled circle. At the bottom, there are four buttons: 'Cancel', '< Back', 'Next >', and 'Finish'. The 'Next >' button is highlighted with a blue border.

In the following screen, you need to provide a name for your forms. Enter the name you want and click **Finish**.



The 'Form Wizard' dialog box is shown with the title 'Form Wizard'. Below the title bar, the text 'What titles do you want for your forms?' is displayed. On the left, there is a preview of a form layout with a yellow background and a grid of fields. On the right, there are two text input fields: 'Form:' and 'Subform:'. The 'Form:' field contains the text 'frmProjects' and the 'Subform:' field contains the text 'frmSubTasks'. Below these fields, there is a message: 'That's all the information the wizard needs to create your form. Do you want to open the form or modify the form's design?'. There are two radio button options: 'Open the form to view or enter information.' and 'Modify the form's design.'. The 'Open the form to view or enter information.' option is selected, indicated by a filled circle. At the bottom, there are four buttons: 'Cancel', '< Back', 'Next >', and 'Finish'. The 'Finish' button is highlighted with a blue border.

Access will give you a preview of what your form looks like. On top, you have the controls on your main form, which is from our **Projects** table. As you go down, you will see a subform. It's like a form within a form.

The screenshot shows the Microsoft Access interface with the 'frmProjects' form open. The form has the following fields:

- Project ID: 37
- Project Name: Cash is King: How to Cut Your Spending by Carrying Cash
- Project Start: 6/10/2013
- Project End: (empty)

Below the form fields is a table named 'frmSubTasks' with the following data:

Task ID	ProjectID	Task Title	Start Date
37	37	Create Outline	6/10/2013
(New)	37	Hire Technical Reviewer	6/10/2013

Multiple Item Form

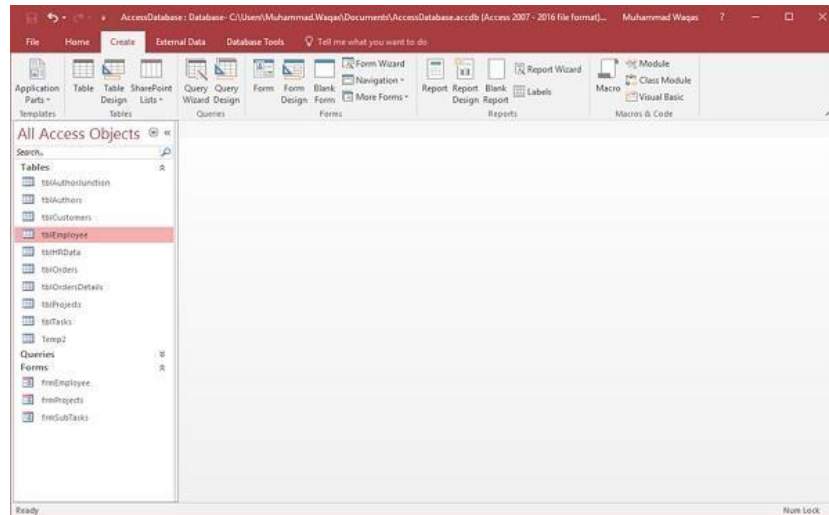
You may also want to create a specific kind of form. For this, you can click on the **More Forms** drop-down menu.

The screenshot shows the Microsoft Access interface with the 'More Forms' dropdown menu open. The menu options are:

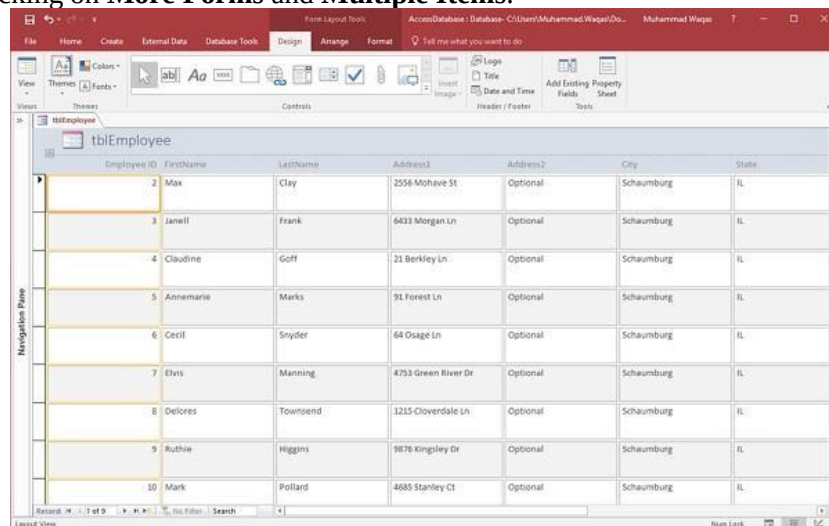
- Multiple Items (highlighted)
- Datasheet
- Split Form
- Modal Dialog

From the menu, you can create a **Multiple Items** form, a **Datasheet** form, a **Split** form, or even a **Modal Dialog** form. These are typically bound forms; select the object that you would like to be bound to that form. This does not apply to the Modal Dialog forms.

To create this type of form, you will need to select the object in navigation pane first. Let us select **tblEmployees** here.



Proceed by clicking on **More Forms** and **Multiple Items**.



The above step will further create a Multiple Items form, listing out all the employees.

Split Form

This type of form is divided in equal halves, either vertically or horizontally. One half displays a single item or record, and the other half displays a list or a datasheet view of multiple records from the underlying data source.

Let us now select **tblEmployees** in the navigation pane and then on **Create** tab. Select **Split Form** option from More Forms menu and you will see the following form in which the form is divided vertically.

File Home Create External Data Database Tools Design Arrange Format Tell me what you want to do

View Themes Colors Fonts

Insert Image Logo Title Date and Time Add Existing Fields Property Sheet

Header / Footer Tools

All Access Objects

Search

Tables

- tblCustomer
- tblCustomer
- tblEmployee
- tblOrders
- tblOrdersDetails
- tblProjects
- tblTasks
- tblTask2

Queries

Forms

- frmEmployee
- frmProjects
- frmSubTasks

tblEmployee

Employee ID: 2 State: IL

FirstName: Max Zip: 60194

LastName: Clay Phone: (847) 555-6432

Address1: 2556 Mohave St Phone Type: Home

Address2: Optional Email: rclay@mycompany.com

Employee ID	FirstName	LastName	Address1	Address2	City	State	Zip	Phone	PhoneType
2	Max	Clay	2556 Mohave St	Optional	Schaumburg	IL	60194	(847) 555-6432	Home
3	Janeil	Frank	6433 Morgan Ln	Optional	Schaumburg	IL	60193	(224) 555-6631	Home
4	Claudine	Goff	21 Berkley Ln	Optional	Schaumburg	IL	60195	(312) 555-3795	Home
5	Annamarie	Marks	91 Forest Ln	Optional	Schaumburg	IL	60193	(224) 555-1111	Cell
6	Cecil	Snyder	64 Osage Ln	Optional	Schaumburg	IL	60194	(224) 555-2123	Cell
7	Elvis	Manning	4753 Green Riv	Optional	Schaumburg	IL	60193	(224) 555-8255	Cell
8	Delores	Townsend	1215 Cloverdal	Optional	Schaumburg	IL	60194	(224) 555-1366	Cell
9	Ruthie	Higgins	9876 Kingsley L	Optional	Schaumburg	IL	60193	(224) 555-4455	Cell
10	Mark	Pollard	4685 Stanley Ct	Optional	Schaumburg	IL	60194	(224) 555-9876	Home
(New)									

Records: 10 of 9

Run Lock